

Council's Best Practice Standards

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The best practice standards from each fact sheet are repeated here for easy reference. The sustainable design categories have undergone some renaming since the original pdf fact sheet suite was published.

Operational Energy

- Achieve an energy efficiency standard at least 10% above minimum NCC compliance.
- Install energy efficient lighting to achieve an improvement of at least 10% above NCC compliance for lighting power density.
- Provide daylight and occupancy sensors for external lighting.
- Provide on-site renewable energy generation.
- Provide external clothes drying facilities.
- Install energy efficient hot water, heating and cooling systems within one energy star rating of the best available.
- Ensure basement carpark are either fully naturally ventilated, or use Carbon Monoxide monitoring to control the speed and operation of ventilation fans.
- Undertake preliminary building energy ratings (e.g. NatHERS, JV3 etc)
- Provide individual utility meters for all dwellings / non-residential areas.
- Provide sub-metering for all major common area services.
- For large developments, commit to regular fine-tuning of building services and their ongoing maintenance to ensure a building's maximum environmental performance.
- For large developments, engage ESD professional(s) on the project design team to provide advice from preliminary design through to construction.

Developments, which seek to vary from these best practice standards, must demonstrate how energy efficiency is satisfactorily achieved.

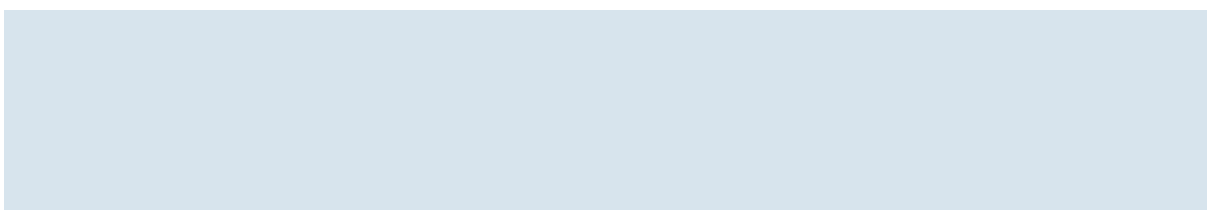
These standards were originally published on the following Sustainable Design Fact Sheets:

- Energy Efficiency
- Construction & Building Management

Transport

- For residential developments, provide at least one secure bicycle parking space per dwelling and one visitor bicycle parking space per 4 dwellings.
- For non-residential developments, provide at least one secure bicycle parking space for 10% of building occupants and sufficient end of trip facilities (showers and lockers).
- For large developments, commit to the development of a Green Travel Plan.
- Incorporate electric vehicle charging infrastructure into the development.
- Allocate 5% or at least 5 parking spaces for motorbikes and/or small vehicles.

Developments, which seek to vary from these best practice standards, must demonstrate how sustainable transport modes will be satisfactorily promoted.



Integrated Water Management

- Install WELS rated fittings within one Star of best available.
- Install rainwater tanks sized to capture the vast majority of otherwise discharged rainwater and plumb to a consistent year round water use such as toilet flushing.
- Install separate water meters in individual dwellings and non-residential areas within the same development.
- Reduce water use for landscape irrigation by selecting low water consuming plants.
- Incorporate measures to reduce water consumption for fire safety system testing, such as re-using water on site and fitting isolation valves or shut off points for sprinkler systems on each floor.
- Exceed Victoria's best practice stormwater performance targets, set out in the Urban Stormwater Best Practice Environmental Management Guidelines (BPEMG) and local integrated water management plans. These targets may be demonstrated by a STORM rating of at least 100% or equivalent modelling results (MUSIC, Urban Developer).
- Prepare a stormwater pollution reduction strategy for the building construction works.

Note: Rainwater tanks sized for retention are in addition to any required for detention in storm events.

Developments, which seek to vary from these best practice standards, must demonstrate how integrated water management can be satisfactorily achieved.

These standards were originally published on the following Sustainable Design Fact Sheets:

- Water Efficiency
- Stormwater Management
- Construction & Building Management

Urban Ecology

- Protect existing on-site vegetation.
- Provide additional vegetation that serve the amenity and environmental performance of the development.
- Provide areas for social interaction between building occupants.
- Incorporate indigenous and/or productive gardens into the design.
- Provide a tap for irrigation and drainage on balconies and courtyards.

Developments, which seek to vary from these best practice standards, must demonstrate how the ecological value of the site is improved.

Indoor Environment Quality

Daylight

- Habitable rooms of single-aspect apartments should be limited in depth to 8 metres from a window.
- Achieve a minimum daylight factor of 1% for 90% of the floor area in each living area including kitchens, and a minimum daylight factor of 0.5% for 90% of the floor area for each bedroom. A daylight modelling report for large scale developments may be required. Please refer to Daylight modelling for planning permit applications for further information.
- For non-residential developments, achieve a daylight factor of at least 2.0% for at least 30% of the floor area of regularly occupied primary spaces.

- Design living areas and private open spaces so that at least 70% of apartments in a development receive a minimum of three hours direct sunlight between 9am and 3pm in mid winter.
- Limit the number of apartments with internal bedrooms to a maximum of 10% of the total apartments proposed.
- Daylight modelling requirements:
 - Model adjacent buildings per their likely developed form based on current zoning (mirrored)
 - Show full extent of internal room modelled
 - Use the following standard reflectance values:
 - Ground plane: 0.1
 - External walls and obstructions: 0.4
 - Floor: 0.3
 - Wall: 0.7
 - Ceiling: 0.8
 - Minimum light transmission values for proposed glazing must be outlined and in line with the proposed design.
 - The grid should contain evenly distributed grid points with a maximum distance of 0.5m.
 - Assume a uniform design sky of 10,000 lux. External obstructions in the form of surrounding buildings, balconies, screening and large trees must be included.

Ventilation

- Design for all dwellings to be effectively naturally ventilated, either via cross ventilation, single-sided ventilation or a combination.
- Design for at least 60% of a development's dwellings to be cross ventilated.
- For adequate single-sided ventilation,
 - Room depth to be no more than 5m.
 - Separated openings high and low or split across the width of the room/ façade (each 5% of the floor area) are preferred
- For adequate cross-ventilation:
 - A breeze path between 2 ventilation openings either within the room or from one room to another.
 - Breeze path length less than 15m measured between ventilation openings and around internal walls, obstructions & partitions.
 - Ventilation openings located either in opposite or adjacent external walls or an external wall and an operable skylight.
 - Size of ventilation openings greater than 2% of total floor area or 1m², whichever is greater. The opening is the maximum allowable clear open area for the window. i.e. if floor area of room is 55m² then clear open area of window / door must be at least 1.1m².
 - No more than 1 doorway or opening between ventilation openings
 - If relying on a courtyard adjacent to a ventilation opening the courtyard must have a minimum depth of 3m from the window or be a minimum size of 9m²
- For adequate mechanically assisted natural ventilation:
 - Deliver fresh air rates between 2.5 - 5 L/s/m² (results should be supported by calculations)

Other

- Locate external noise sources to prevent acoustic impacts to openable windows.
- Design external shading devices to provide protection from summer sun angles and respond to different façade orientations.
- Provide openable external windows to circulation corridors and lift lobbies to facilitate natural ventilation and daylight.
- Avoid the use of light courts for daylight provision to habitable rooms.

Developments, which seek to vary from these best practice standards, must demonstrate how indoor environment qualities can be satisfactorily achieved.

Waste & Resource Recovery

- Adopt a recycling target of at least 70% for all demolition and construction waste.
- Provide recycling facilities that are at least as convenient for building occupants as general waste facilities.
- Provide on-site management of food and garden waste.
- Commit to the preparation and availability of a Building Users Guide (BUG) and/or Building Operations Manual for large developments.
- For larger developments, engage ESD professional(s) on the project design team to provide advice from preliminary design through to construction

Developments, which seek to vary from these best practice standards, must demonstrate how waste management and resource recovery are satisfactorily addressed.

These standards were originally published on the following Sustainable Design Fact Sheets:

- Waste Management
- Construction & Building Management

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